CYBERSECURITY LOG ANALYZER PROJECT REPORT

1. ABSTRACT

Cybersecurity Log Analyzer V3 is a security monitoring application developed using Python, FastAPI, SQLite, and SQLAlchemy. The system analyzes security logs, identifies brute-force attack patterns, stores alerts in a database, and provides REST APIs for threat monitoring and management.

2. OBJECTIVES

- Detect suspicious login attempts.
- Analyze security logs automatically.
- Store detected threats in SQLite.
- Provide REST APIs for monitoring alerts.
- Build a foundation for SIEM-like security monitoring.

3. TECHNOLOGIES USED

Programming Language: Python
Framework: FastAPI
Database: SQLite
ORM: SQLAlchemy
API Documentation: Swagger/OpenAPI
Server: Uvicorn

4. SYSTEM ARCHITECTURE

Log File → Analyzer Engine → Threat Detection → SQLite Database → FastAPI APIs → Security Analyst

5. MODULES IMPLEMENTED

• Log Analysis Module
• Brute Force Detection Module
• Database Storage Module
• Alert Management API
• File Upload API
• Swagger Documentation

6. API ENDPOINTS

GET / : Health Check
GET /alerts : Retrieve stored alerts
POST /upload : Upload log files

7. PROJECT WORKFLOW

1. Read security log files.
2. Parse IP addresses.
3. Detect repeated failed login attempts.
4. Store alerts in SQLite.
5. Retrieve alerts using REST APIs.

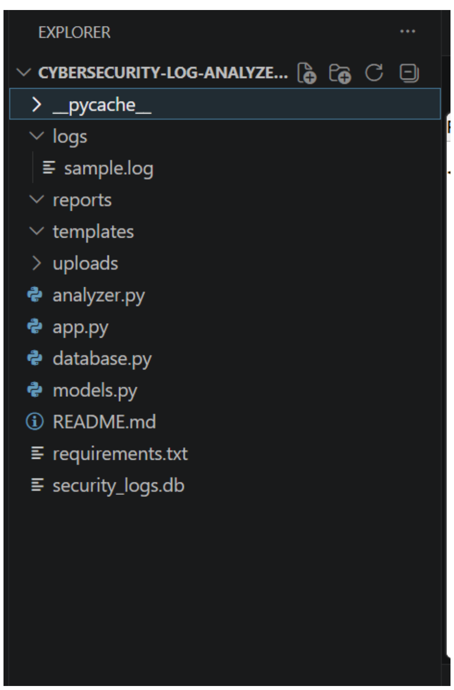
8. RESULTS

The application successfully detected brute-force attacks and stored them in SQLite. Alerts were accessed through FastAPI endpoints and validated using Swagger UI.

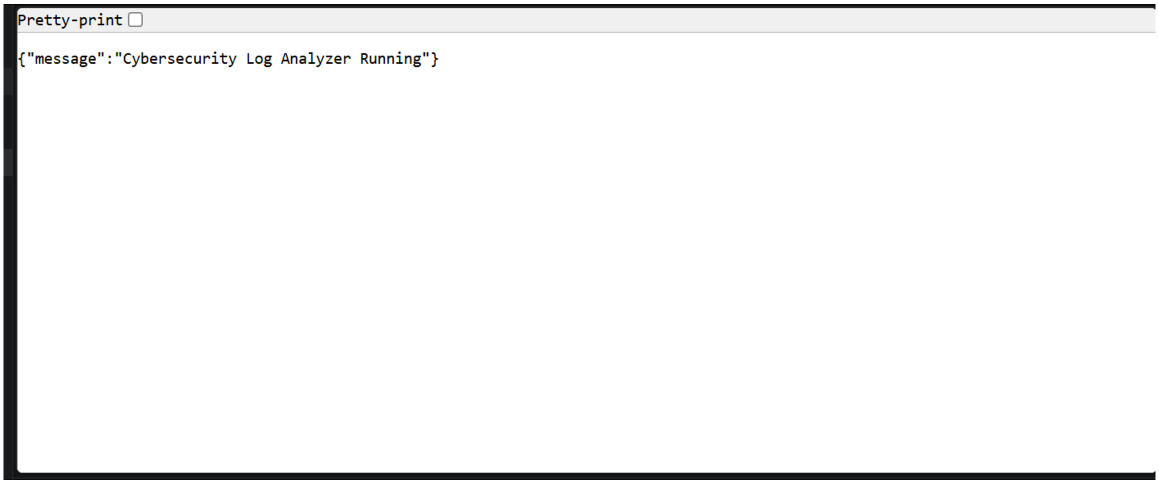
9. SCREENSHOTS

Insert screenshots below each heading:

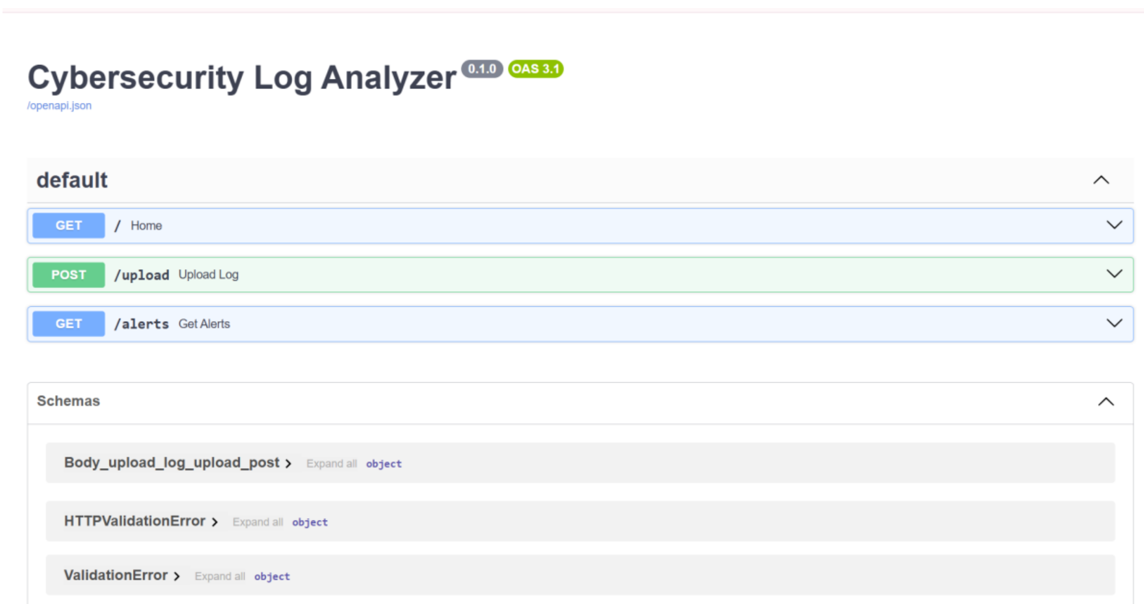
9.1 Project Structure Screenshot



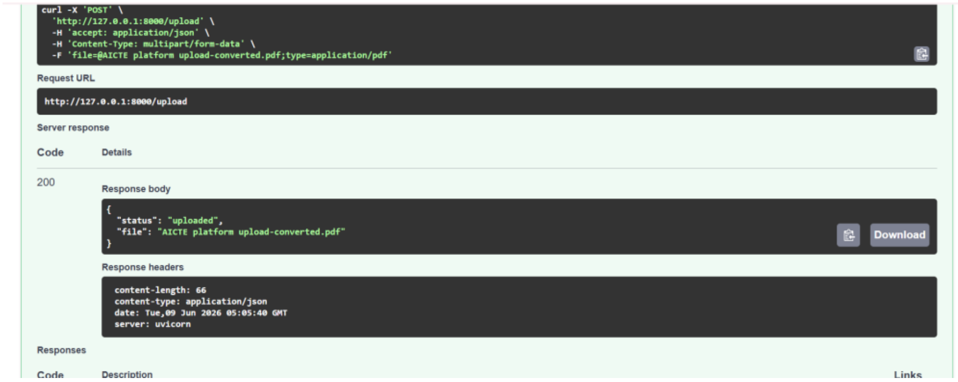
9.2 Home API Response Screenshot



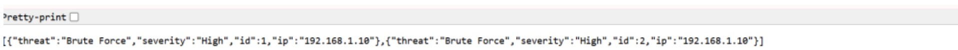
9.3 Swagger Documentation Screenshot



9.4 Upload API Success Screenshot



9.5 Alerts API Output Screenshot



10. ADVANTAGES

- Lightweight and easy to deploy
- Demonstrates practical cybersecurity concepts
- Supports API-based monitoring
- Extensible for advanced threat detection

11. FUTURE ENHANCEMENTS

- SQL Injection Detection
- XSS Detection
- Real-time Monitoring
- Dashboard using Plotly
- JWT Authentication
- Docker Deployment

12. CONCLUSION

Cybersecurity Log Analyzer V3 demonstrates the practical implementation of log analysis, threat detection, API development, and database integration. The project serves as a strong foundation for advanced cybersecurity monitoring solutions.